ezLCD-5035-RT Manual
&
Quick Start Guide

May 2024

Foreword

Thank you for purchasing an ezLCD! In 30 Years of providing TFT LCD touchscreen solutions and 20 years since the introduction of the first ezLCD-001, this is the first time we can truly say the ezLCD can be your ENTIRE product. Of course, it can be your HMI (human user interface) tied to a different controller solution. With the evolution of semiconductor technology this is the first model of the ezLCD-5x family has enough power to be a complete PLC, touchscreen control system, thin client, or stand-alone instrument. The power of the STM32H757XIH6 microprocessor harnesses powerful graphics and I/O in a solution, drawing as little as one watt of power! I do not have the space to name all of the team members, customers and consultants who have contributed to making this evolution of the ezLCD family possible. I do owe a shout out to Michael S. who started the product line and was its founding father for 8 years until his passing to the great lab in the sky. Also, Robert G. and Jacob C. were instrumental in the creation and release of this product generation. A requirement of every EarthLCD employee to spend time programming an ezLCD has helped us put the ‘ez’ in ezLCD. The many suggestions (and patience) from beta testers, including Jim C, have been invaluable in making this a better product. We hope you enjoy using our product and are open to your suggestions for continuing to improve it. Be sure to check the product page for the ezLCD-5035-RT for updates!

Randy from the Big Easy!
Randy@EarthLCD.com

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1. Using the Documentation

The Documentation used for Configuring and Programming your ezLCD is:

1.1 "ezLCD-5035-RT Manual" (This Document!)

May be specific for a particular model or family (ezLCD-5035)
Provides "Quick Start" instructions.
Describes how to connect and verify operation of your ezLCD.
Describes the options for using the ezLCD.
Describes how to load new firmware, and how to customize your ezLCD device.

1.2 "ezLCD API Manual"

Common for all ezLCD-5035-RT products. Describes the set of commands, which can be sent to the ezLCD through any of the implemented interfaces (USB, RS232, SPI, Wi-Fi etc.). Those commands may be sent by an external host (PC or microcontroller) or by a scripting language such as Lua running on the ezLCD.

1.3 "ezLCD Lua API Manual"

Common for all ezLCD-5035-RT products. All ezLCD-5035-RTproducts have an embedded Lua interpreter. ezLCD Lua has been developed to access all graphic and I/O capabilities of the ezLCD API using the Lua language. These commands are the Lua commands that are specific to the ezLCD and will start with “ez.”, such as ez.Cls(). You can use the resource below to learn the standard Lua commands.

“Programming in Lua (Fourth edition) By Roberto Lerusalimschy”**
The official book about the Lua programming language written by the creator of the language and describes the general commands. It is available at:

http://www.amazon.com/exec/obidos/ASIN/8590379825/lua-docs-20

The original edition is free to read online here:
https://www.lua.org/pil/contents.html .

In the \lua directory on your ezLCD there are many examples. “learn.lua” shows the most common commands. blink.lua shows how to access an ezLCD I/O signal. demo.lua shows many graphic commands.

More information about Lua can be found at: *http://www.lua.org/*

Of course, there are many videos on YouTube for learning Lua.

**Not included. Must be downloaded or purchased separately.

1.4 “Your ezLCD Flash Drive”

When you plug in your ezLCD to the USB port of your computer, the ezLCD Flash Drive will appear in your computer’s file system as a disk drive and will be here after referred to as the ezLCD Flash Drive.

[[*Note the ezLCD Flash Drive can be the SD card or the 16 Mbyte QSPI flash chip on the EarthSOM-STM32H7 depending on the firmware.* **]]**

The ezLCD Flash Drive, when shipped from the factory, will have all the documentation contained in the ‘Documentation’ directory. The Lua directory on the ezLCD Flash Drive has Lua examples that run on your ezLCD. Classic program examples such as “Hello World” and “Blink” for your ezLCD will help you understand how Lua works on an ezLCD. A link to the latest release of the ezLCD Flash Drive contents can be found on the ezLCD-5035-RTproduct page on our website.

1.5 “Read Me.txt File on the ezLCD Flash Drive”

This file may contain essential information about the latest version of your product’s firmware, and the ezLCD Flash Drive contents.

2. ezLCD-5035-RT Overview

The ezLCD is a Smart LCD designed to add a touchscreen LCD interface to existing products in the easiest, fastest, and most efficient way, or operate as a standalone embedded computer system. The ezLCD is unique in that it is bare metal; not using Linux, Windows or a bloated operating system. This results in instant booting, low power consumption, simplicity, lower cost, and a more reliable system. It consists of an LCD, a touchscreen, and a processor board with input & output interfaces. The goal of the ezLCD product line (since 2004) is to be easy to use by engineers, students, integrators, and/or Makers.

The ezLCD-5035-RT is an all-in-one programmable color LCD panel which includes:

- Color TFT LCD
- Resistive or Capacitive Touchscreen
- A 32-bit STmicro STM32H757XIH6 ARM Microcontroller with an LCD Controller
- SDRAM for the LCD Screen Data
- MicroSD Card slot for storing Lua programs, bitmaps, fonts, user, and config data
- Power supply
- USB port for Programming & Configuring your ezLCD with a Personal Computer

The ezLCD-5035-RTcommunicates with the outside world through several interfaces:

- CANBus*
- RS-232 Serial Port
- TTL Serial Port
- RS-485*
- USB

- I2C
- SPI
- SD/MMC
- Ethernet**

* Requires external driver IC

** Requires external ethernet jack breakout and future firmware release

The ezLCD-5035-RT firmware is in-field updatable and contains an extensive command set callable from the ezLCD API, or the Lua scripting language including:

- Graphic commands
- Both BMP and JPG image support
- Hardware JPEG on ezLCD-5035
- True Type, Open Type Bitmap font rendering
- Unicode support
- SD file I/O (FAT32)
- Touch Screen commands

The ezLCD-5035-RT may be in-field customized by:

- Adding custom fonts
- Adding custom bitmaps in BMP and JPG formats
- Customizing the startup screen
- Modifying interface parameters like RS232 baud rate
- Modifying pin functions

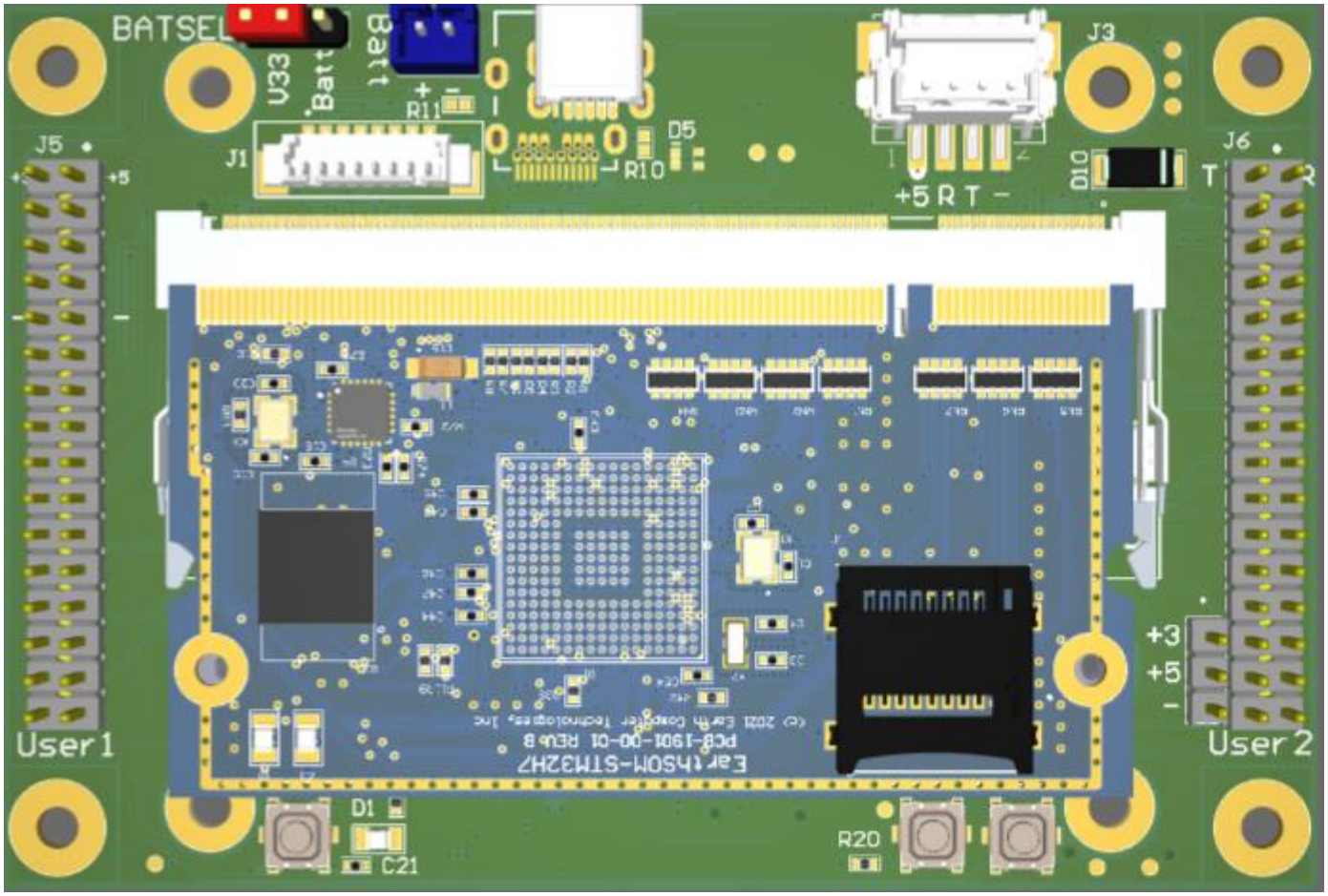
The ezLCD-5035-RT belongs to the ezLCD 5x family of intelligent displays. Note that it is modular based on a SOM (system on module), the EarthSOM-STM32H7. Anything you develop on the ezLCD-5035-RT will be able to run on other family members with sizes up to 15” and capacitive or resistive touch screens. It will also allow EarthLCD to rapidly develop custom sizes and configurations. Please see “Appendix C. ezLCD Hardware and Software Roadmap” near the end of this manual for future planned product and software enhancements.

All ezLCD-5x devices (including the ezLCD-5035-RT) can be used in multiple operating modes:

1. ezLCD API - The ezLCD API Commands allow easy access of all the hardware capability by an external processor, or internal languages of the ezLCD described in the ezLCD API Manual. The ezLCD is driven by a set of commands, which can be received through any of the implemented interfaces such as serial, CANBus*, and SPI*. The ezLCD-5x API Manual describes those commands in detail. *Under development.
2. Lua – Lua programming language. The ezLCD holds an embedded Lua programming language interpreter. The ezLCD Lua library is described in the *ezLCD Lua API Manual*. The base Lua commands are described well online & particularly in the Lua PIL book.
3. Arduino Mode – ***Upcoming Feature*** – By using the ezLCD Arduino Library 'arLib', which uses the ezLCD API, you will be able to use the Arduino IDE with the ezLCD Arduino Library on various microcontroller boards including the Arduino Uno, ESP32, and Raspberry Pi Pico.

The ezLCD may be used as an "intelligent" display (HMI) with a host processor using it as user interface peripheral as a stand-alone embedded touchscreen system. There is enough flash memory left to incorporate added graphical instructions, or to customize the software for tasks. The ezLCD API simplifies complex peripherals & protocols into single line API calls. Furthermore, with the Lua scripting language API, the ezLCD becomes a powerful and easy to use embedded computer. This makes the ezLCD ideal for rapid product development and prototyping. Applications include automotive, avionics, marine, industrial control, instrumentation, hobby, etc.

3. Design Concept



The ezLCD-5035-RTconsists of:

- **The EarthSOM-STM32H7**
  - ST Microelectronics STM32F7 32-bit ARM Cortex-M7 processor
  - 2MB Fast internal SRAM
  - 16 MB SDRAM
  - Micro SD Card allowing up to 32 GB customer supplied storage
  - 16 MB SPI Flash Drive to be used as an alternative to Micro SD
  - Ethernet PHY*
  - Embedded TTL & DSI LCD controller supporting up to HD displays
  - Hardware JPEG and motion JPEG Video Playback

  - Rich peripheral set up to GPIO, I2C, SPI, UART, ADC, DAC, PWM, and CAN
  - ezLCD Smart LCD, API, and Lua scripting language “instant on” with NO LINUX operating system.
  - 10 Year Availability

* New Feature Support Coming

- **Personality Board**
  - Interfaces the EarthSOM with the outside world. It contains a multitude of I/O interfaces, a DC/DC voltage regulator, touchscreen controller circuitry, an LCD backlight driver and the LCD connector. On the ezLCD-5035-RT the personality board also serves as the mounting bracket for the ezLCD, and has the mounting holes for the LCD panel standoffs.
- **LCD Panel**
  - EarthLCD-3.5-320240-RT uses a 500 nit TFT IPS display. EarthLCD was the first TFT LCD distributor in the United States and has a history of providing LCDs with long term availability for over 30 years.

- **LCD Backlight Inverter**

The EarthSOM is connected to the Personality Board by a SODIMM connector, which is like a notebook memory module connector. It also has two standoffs with screws to provide a vibration resistant connection. Upon power-up, the EarthSOM reads the ezLCD parameters and other configuration data from the firmware in the STM32H757 microcontroller.

The design concept described above enables the firmware and the EarthSOM to be identical for several types of LCD graphic displays available on the market today. The LCD controller embedded in the EarthSOM is capable of driving TFT, STN, Color, or Monochrome LCD displays with a resolution of up to 1024x768.

Incorporating a new LCD panel would require the following:

1. Designing a Personality Board for the particular LCD panel.
2. Modifying the ezLCD firmware to support the LCD timing and Touch Screen Controller and any peripherals added.

EarthLCD can design custom personality boards for different LCD sizes up to 15”, with or without resistive or capacitive touchscreens. We have several ezLCD’s under development currently. Contact EarthLCD sales for more information. Also look at “Appendix C. ezLCD Hardware & Software Roadmap”.

4. Quick Start

For those of you who could not wait to read everything in this manual before starting to use it, please read it later! You can also go to the EarthLCD channel on YouTube and watch the quick start video for the ezLCD-5035-RT.

Plug it in!

- Plug in a USB cable from the ezLCD to your computer.
- The ezLCD Startup Screen will come up showing the product model at the top of the screen and the firmware version at the bottom of the screen. Also, a QR code will display bottom left. This is the product/CPU serial number.
- A USB drive will pop on your computer. Copy the contents of the ezLCD Flash Drive directory to your PC as a backup.
- At this point, if you plan on using only the ezLCD API in your project, you can skip to 4.3 Installing and Using the ezLCD Utility. To see the full capability, you still may choose to go through the 4.1 Lua Quick Start, as it will allow you to prototype your projects visual appearance quicker and experience the features of the ezLCD from the many examples in the Lua scripts directory provided on the ezLCD Flash Drive.

Usertest.lua

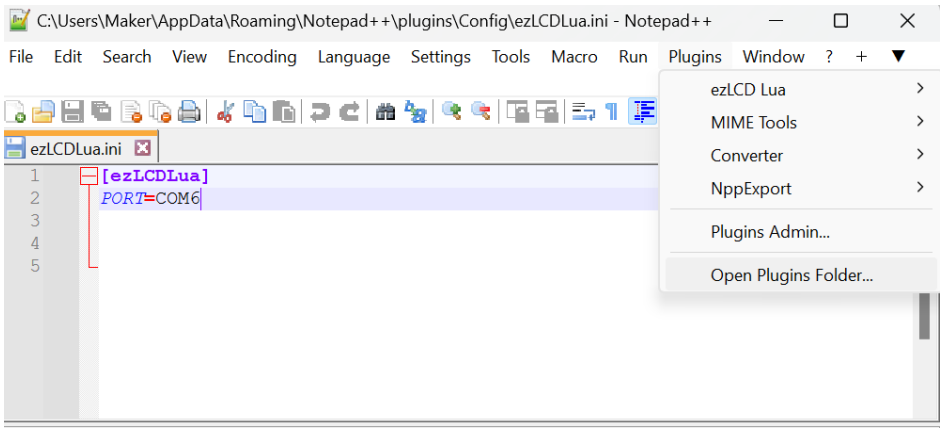
At the ezLCD factory we use a program called usertest.lua to do the final production test for the ezLCD. This program can be accessed from the Boot Options menu. If you press & hold the B1 button on the rear of the ezLCD and then tap reset, the ezLCD Boot Options screen will come up. If usertest.lua is on the ezLCD Flash drive as shipped from the factory, the 'Run User Test' option button will show on the screen. When you press it, the program will run, giving you the option of calibrating the touchscreen or running the test/demo script. Touch the screen to step through the different sections of the program. At the end it will optionally rename autoexec.lua to usertest.lua. Note for your final product you can put your own usertest.lua script on the ezLCD to demo or setup/configure your product. Read more about this feature in the Boot Options section of the manual.

About the Command Port

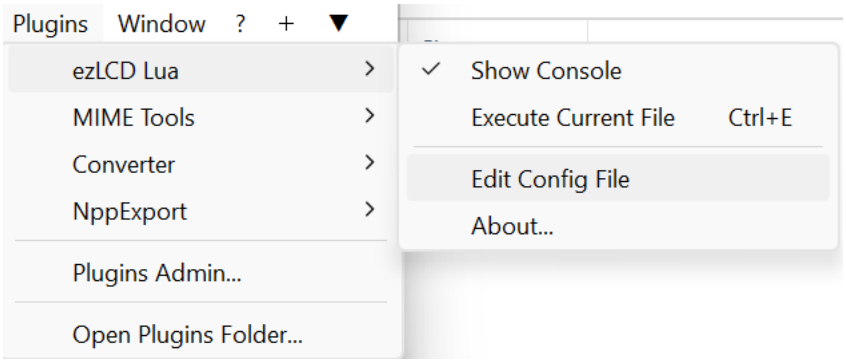
The ezLCD is really just a touchscreen computer designed for embedded applications. It is optimized to be an easy-to-use standalone touchscreen embedded computer system, or a peripheral to another system (referred to as the host). The power of the ezLCD is an extensive API, which simplifies complicated I/O interfaces into simple to use peripherals, most of which can be configured or accessed with simple to read one-line commands! These commands are sent to the ezLCD through a Command Port which can be a serial (TTL or RS-232), or USB. The USB port is intended for development, and we created editor/IDE plugins to use with Notepad++ or Microsoft Visual Studio Code. It is possible of course to have a host computer use the ezLCD as a GUI client through the command port (and all the I/O ports as intelligent peripherals). There are many ways to use the command port for development, but we created a utility for learning the commands called the ezLCD config Utility; a Notepad++ plugin for using this popular editor for developing Lua applications, and ezLCD Studio to program & manage your ezLCD project workflow. ezLCD Studio is written in Microsoft Visual Studio Code.

4.1 Lua QuickStart Using Notepad++

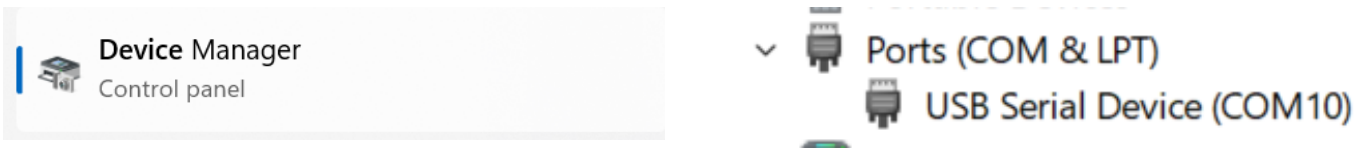
1. Download Notepad++ from the official website and install.
2. In Notepad++ use the menu option, Plugins > Open Plugins Folder.



3. Create a folder named ezLCDLua.
4. Copy the ezLCDLua.dll file provided on your ezLCD from the Notepad++ folder into the ezLCDLua folder.
5. Close then reopen Notepad++.
6. Use the menu option, Plugins > ezLCD Lua > Edit Config File.

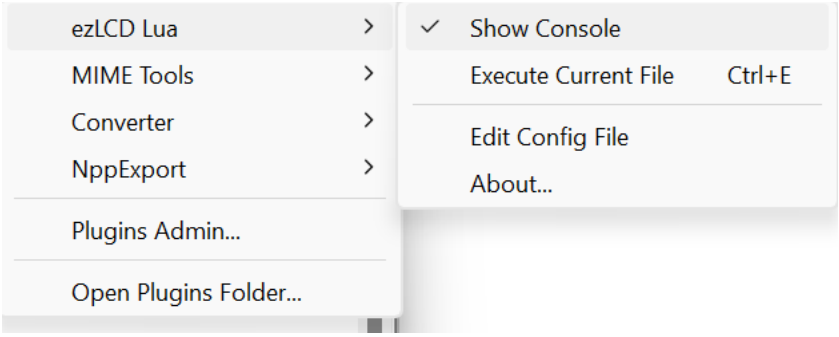


7. Using Windows 10 "Device Manager" click on "Ports (COM & LPT)". You should see a line that says USB Serial Device (COMxx) where xx is a number.



8. In the config file replace COM1 with the number you see in Device Manager and SAVE the file.

9. Use the menu option Plugin > ezLCD Lua > Show Console.



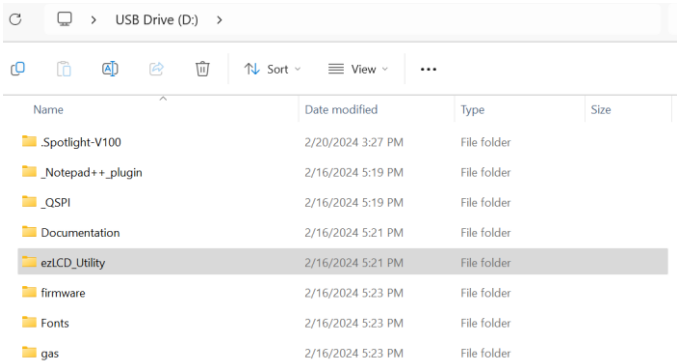
10. Start typing commands like ez.Cls(ez.RGB(255, 0, 0)) in the console, and enjoy.
11. You can also run Lua files by opening a .lua file in Notepad++ and pressing CTRL + e.
12. For better workflow performance and to build your library on your host, we suggest you create a work folder and edit your lua files on your computer. When you are ready to save it to the ezLCD for use, select the File>Save A Copy As menu option in Notepad++.
13. We encourage you to run all the scripts in the lua directory to become familiar with their capability.
14. When your project is complete, you can simply name your main lua script autoexec.lua and copy it into the root of the ezLCD Flash Drive. To keep autoexec.lua from running you can hold the B1 button on the back of the ezLCD and press then release reset. This will bring up the boot options menu where you can press the “No Autoexec” button followed by the “Close” button.

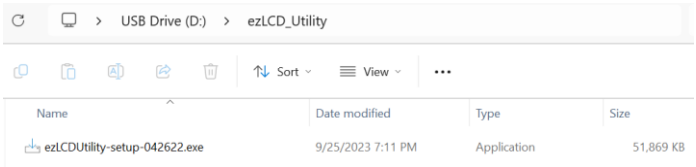
4.2 Visual Studio Code QuickStart

1. Open an empty project folder on your computer to create a new project folder.
    - File -> Open Folder
2. Create a folder called "sd" in the root of your project. This allows you to segregate design files from files that need to be transferred to your display.
3. Create any additional design folders in the project. These will be in the root of your project and not uploaded to your display.
4. Plug in your ezLCD-5035-RT to the USB port of your computer.
5. In the LCD View do the following (right-click on TreeView) to:
    - Set LCD Root File System to your ezLCD Flash Drive letter
    - Set Serial Port
    - Test Serial Port (The screen should turn off and on)
6. In the file explorer, ‘right click’ on files in your "sd" and select "ezLCD Upload File(s)" to upload them to your display.
7. To run programs from your project folder:
    - Select explorer in the left task bar
    - Select a program
    - Right click program name in project folder and select ezLCD save all, upload and run.
    - To run a program without saving, right click and select ezLCD, then ezLCD Run File on LCD.

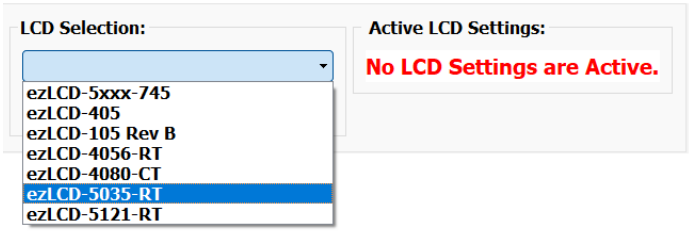
4.3 Installing and using the ezLCD Utility

1. Install the ezLCD Utility by running the .exe file in the ezLCD_Utility directory. Once installed, run the ezLCD4xx_Config_Utility from your personal computer



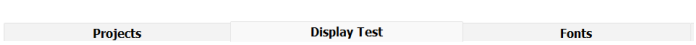


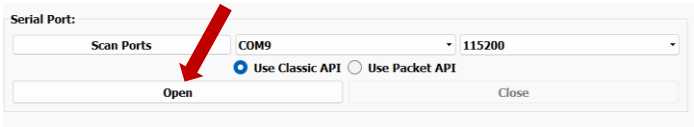
2. In the LCD selection pull down select your ezLCD Model then click the APPLY Button





3. Click DISPLAY TEST then click SCAN PORTS and click OPEN





- The bottom of the screen will display connected to COM1 (1=whatever USB port the ezLCD is at)

  EX: Connected to COM49 : 115200, 8, 0, 1, 0

4. In the Projects Tab, click Lua on the left menu ribbon. Next to the RUN_LUA_SD, type “lua/hello.lua” and then click the RUN_LUA_SD button. The screen should turn blue and print "Hello World!!!" in gold.

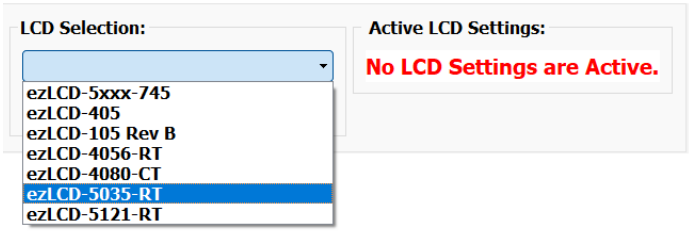


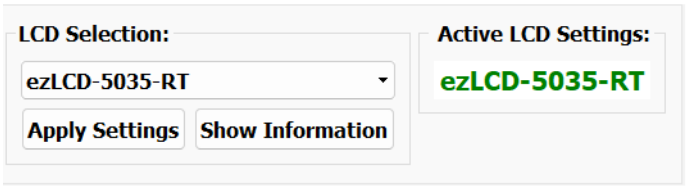
- Note: You can also open the script with your editor and paste in the "Run Lua commands over API" screen below and click the RUN_LUA button to run it. Also, you can do edits in there and re-run it. It is suggested you reset the ezLCD after every script. You will need to go back to the Connection Menu to CLOSE then re-OPEN the port to reconnect it.

4.4 Firmware Upgrade

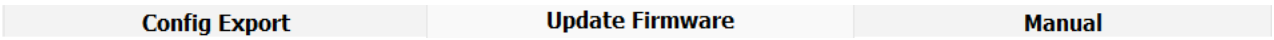
Your ezLCD ships from the factory with the latest version, but if an upgrade is released this is how it is installed.

1. Launch the ezLCD Utility
2. In the LCD selection, pull down to select the ezLCD model you have, then then click the APPLY Button.





3. Click on the “Update Firmware” Tab.



4. Click on the “OPEN” Button.
5. Navigate to the ezLCD firmware file on your PC (ezLCDnnnn-xxxx.dfu) and select it.



6. Plug in your ezLCD to the USB.
7. Press & hold the “RESET” Button while also Pressing & Holding the “BOOT0” Button (NOTE: 10x/Proto models hold BOOT button down next to power switch while powering on unit). The screen should display no date and both the LED on the SOM and on the ezLCD should be off to signify the ezLCD is in bootloader mode.
8. Release the “RESET” Button, then release the “Boot 0” Button
9. Click on the “UPDATE” Button

5. ezLCD Customization

5.1 User Configuration (config.txt)

The config.txt file is used to modify some of the ezLCD-5035-RT default parameters like:

- Communication parameters (RS232 baud rate, parity, IP address, etc.)
- Startup screen
- Startup backlight
- Pin functions

Supported File Types:

File Type	Supported Extensions
Bitmaps (Icons)	.bmp, .jpg, .ezp
Bitmap Fonts	.ezf
Free Type (True Type) Fonts	.ttf
Scripts	.lua

Upon power-up or reset, the ezLCD-5x CPU configures itself according to the user configuration keys read from the config.txt file. The concept of config.txt is similar to the one used in the Windows .ini files. It consists of the set of keys and values assigned to them. The keys are described in the section: Configuration Keys

An index for each of the sections [Fonts]/[Images] must start at 0 and increase by one for each new entry in the section. Indices cannot be skipped. This index is used as an argument in ezLCD-5x Commands and in the Lua instructions. The indices must be assigned in the order of the files of the particular type that appear in config.txt file.

config.txt Syntax

- config.txt is an ASCII text file.
- Each text line is terminated either with <CR><LF> (DOS, Windows, etc.) or the <LF> alone (UNIX, Linux, MAC OS X, etc.).
- Comments can start with '#' or ';'
- Each text line can contain only one key assignment.
- Keys and their values are separated by '='.
- Keys, and their values, are case insensitive.

- Values maybe specified in the following formats:
  - Decimal Numbers
  - Special values: Even, Odd, hw, h/w, sw, s/w, Xon/Xoff, USB, Rs232, SPI, I2C, Net, Mac, Ethernet, ezButton, cuButton, CalibratedXY.

Example of the config.txt file contents:

```
; ezLCD5035 Config.txt 04/19/2024
;
[Startup]
Backlight = On
Brightness = 80
StartScreen = Default
; TouchScreen = CalibratedXY
;
; Font number must start at 0 and increase by 1 for each additional font
; No numbers can be skipped, parsing stops if there is a gap in the image sequence
[Fonts]
0 = \Fonts\01-OSRegular.ttf
1 = \Fonts\04-PTSerifBold.ttf
2 = \Fonts\06-UnispaceRegular.ttf
3 = \Fonts\11-SteelfishRegular.ttf
4 = \Fonts\14-ComicRegular.ttf
5 = \Fonts\17-GaramondRegular.ttf
6 = \Fonts\25-TahomaBold.ttf
7 = \Fonts\28-VerdanaRegular.ttf

; Image number must start at 0 and increase by 1 for each additional image
; No numbers can be skipped, parsing stops if there is a gap in the image sequence
[Images]
0 = /Images/RedClosed.jpg
1 = /Images/GreenOpen.jpg
2 = /Images/EarthLCD_640x480_Splash.jpg
3 = /Images/bwimage.jpg
4 = /Images/car320.jpg
```

5 = /Images/hotair320.jpg
6 = /Images/image2.jpg
7 = /Images/EarthLCD_320x240_Splash.jpg
8 = /Images/EarthLCD_320x240_Splash.jpeg
9 = /Images/ezlcdicon.jpg
10 = /Images/car320.jpg
11 = /Images/ezlcdicon.jpg
12 = /Images/GreenOpen.jpg
13 = /Images/hotair320.jpg
14 = /Images/image.jpg
15 = /Images/made-in-usa320.jpg
16 = /Images/moon320.jpg
17 = /Images/RedClosed.jpg
18 = /Images/Start.jpg
19 = /Images/ezlcdicon.jpg

Note: If key settings are not present, they revert to default settings shown below.

5.2 Configuration Keys

Section	Key	Setting	Description
Startup	Backlight	On	LCD Backlight is initially on (default)
		Off	LCD Backlight is initially off
	Brightness	0-255	LCD Initial Brightness (100=default)
	SerialBaudRate	300-115200	TTL Serial port baud rate (default=115200)
	TouchScreen	None	Touch Screen protocol initially disabled (default)
		CalibratedXY	Touch Screen protocol initially CalibratedXY
		ezButton	Touch Screen protocol initially ezButton
		cuButton	Touch Screen protocol initially cuButton
	USBCommandLogging	Off	USB Port used for commands
		On	USB Port receives command log via CDC (for debugging)
	ExtendedMode	Off	ezLCD initially starts in standard mode (320x240) (default) *ezLCD-405 Only
		On	ezLCD initially starts in extended mode (640x480) *ezLCD-405 Only
Startup	APIMode	Classic	Serial API is not packet-based (ie, no CRC checking). Compatible with legacy products
		Packet	Serial API Commands encapsulated in CRC-verified packets with ACK/NAK Responses
	StartScreen	Default	ezLCD starts with blank screen showing LCD model and firmware version (default)
		Image	ezLCD starts displaying selected image
		Color	ezLCD starts with blank screen of selected color

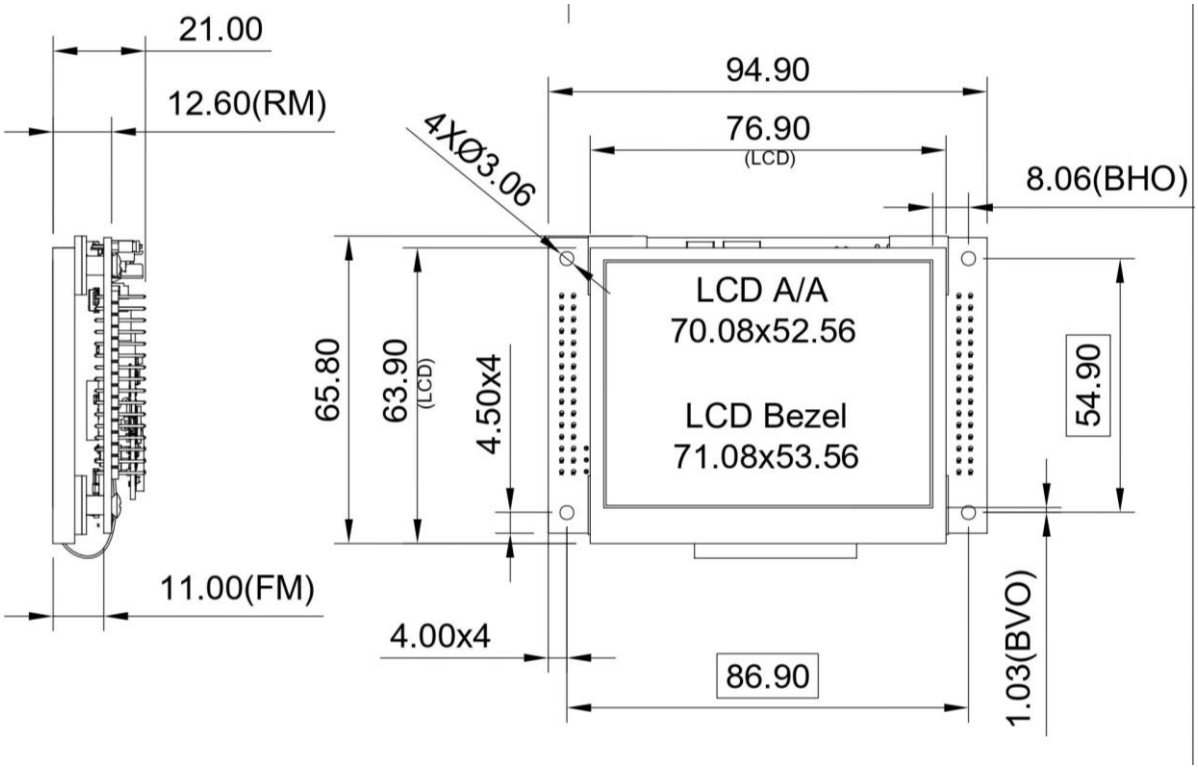
		Slideshow	ezLCD sequentially displays images in numeric order after boot
		FactoryTest	Enable Factory Test mode on startup
	ImageNumber	0-99	Selects the image to be displayed on boot, if StartScreen Image mode was selected
	ImageX	0-639	Upper left X coordinate for displayed image if StartScreen image or Slideshow mode was selected
	ImageY	0-479	Upper left Y coordinate for displayed image if StartScreen image or Slideshow mode was selected
	SSColor	0-65535	Color to be used for the blank screen if StartScreen Color was selected
	SlideshowDelay	0-255	Delay in seconds between Slideshow images (default=5)

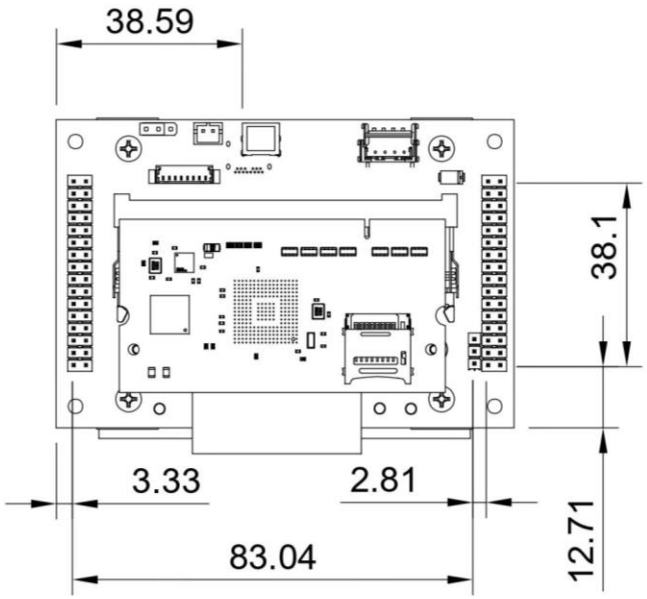
	Transmitter	Serial	EZLCD_READY sent to Serial port after boot (default)
		USB	EZLCD_READY sent to USB port after boot
		None	EZLCD_READY not sent after boot
Fonts	0	File Path	Path (on SD Card) of Font 0 EZF file
	…		
	{n}	File Path	Path (on SD Card) of Font n EZF File
Images	0	File Path	Path (on SD Card) of Image 0 BMP, EZP, or EZH file to be loaded in cache on boot
	…		
	{n}	File Path	Path (on SD Card) of Image n BMP, EZP, or EZH file to be loaded in cache on boot
	Decompress0	Yes	Image 0 will be decompressed during load to cache

	…		
	Decompress{n}	Yes	Image n will be decompressed during load to cache

6. Hardware and Interfaces

6.1 Dimensions

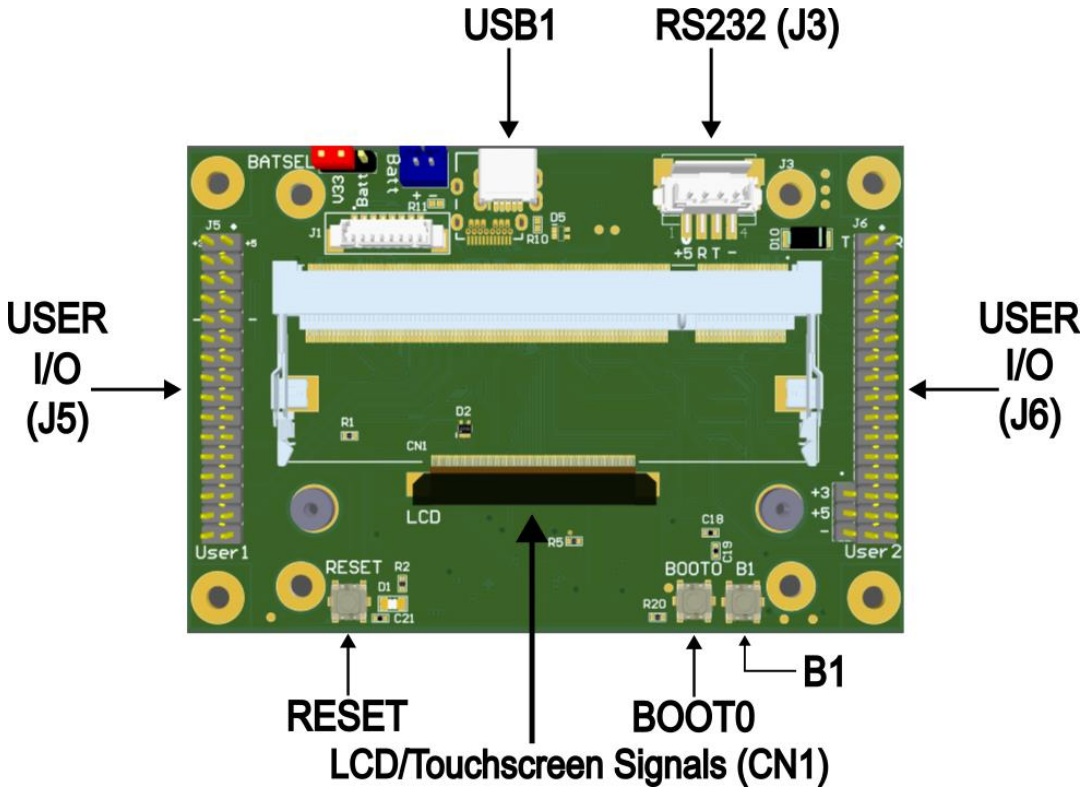




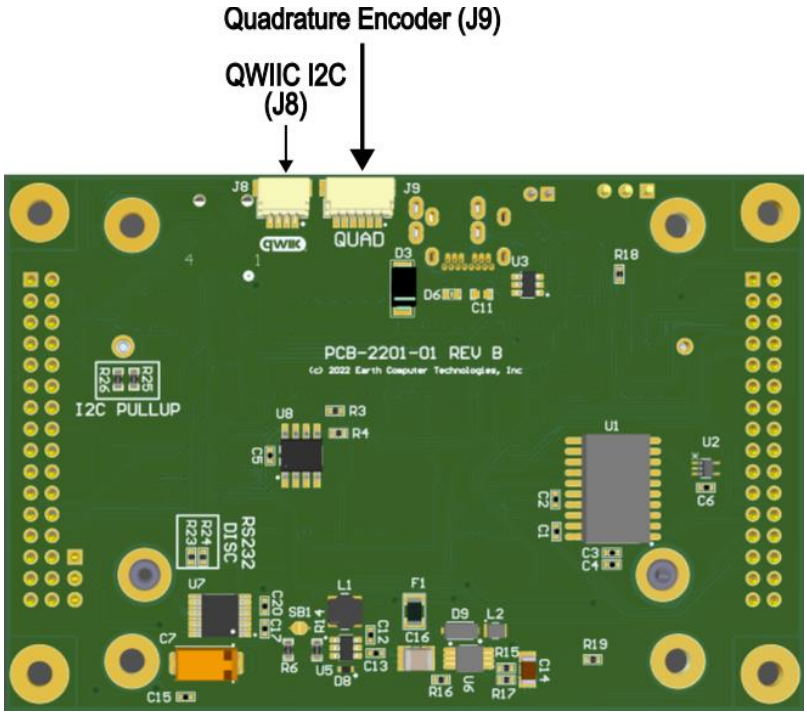
Legend	
BHO	Bezel Horizontal Offset
BVO	Bezel Vertical Offset
FM	Front Mount
RM	Rear Mount

6.2 Connector Locations

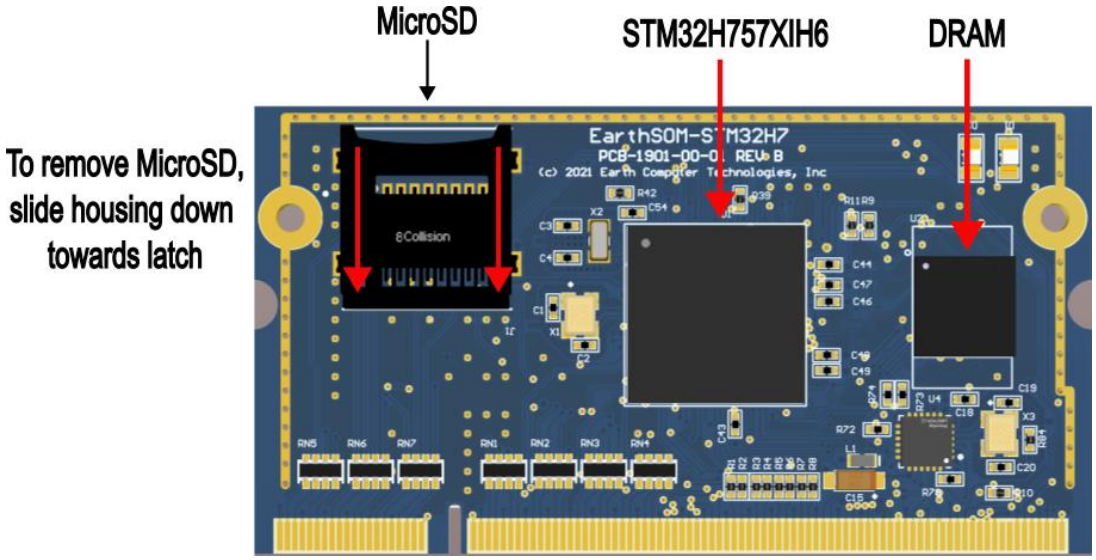
PCB Front



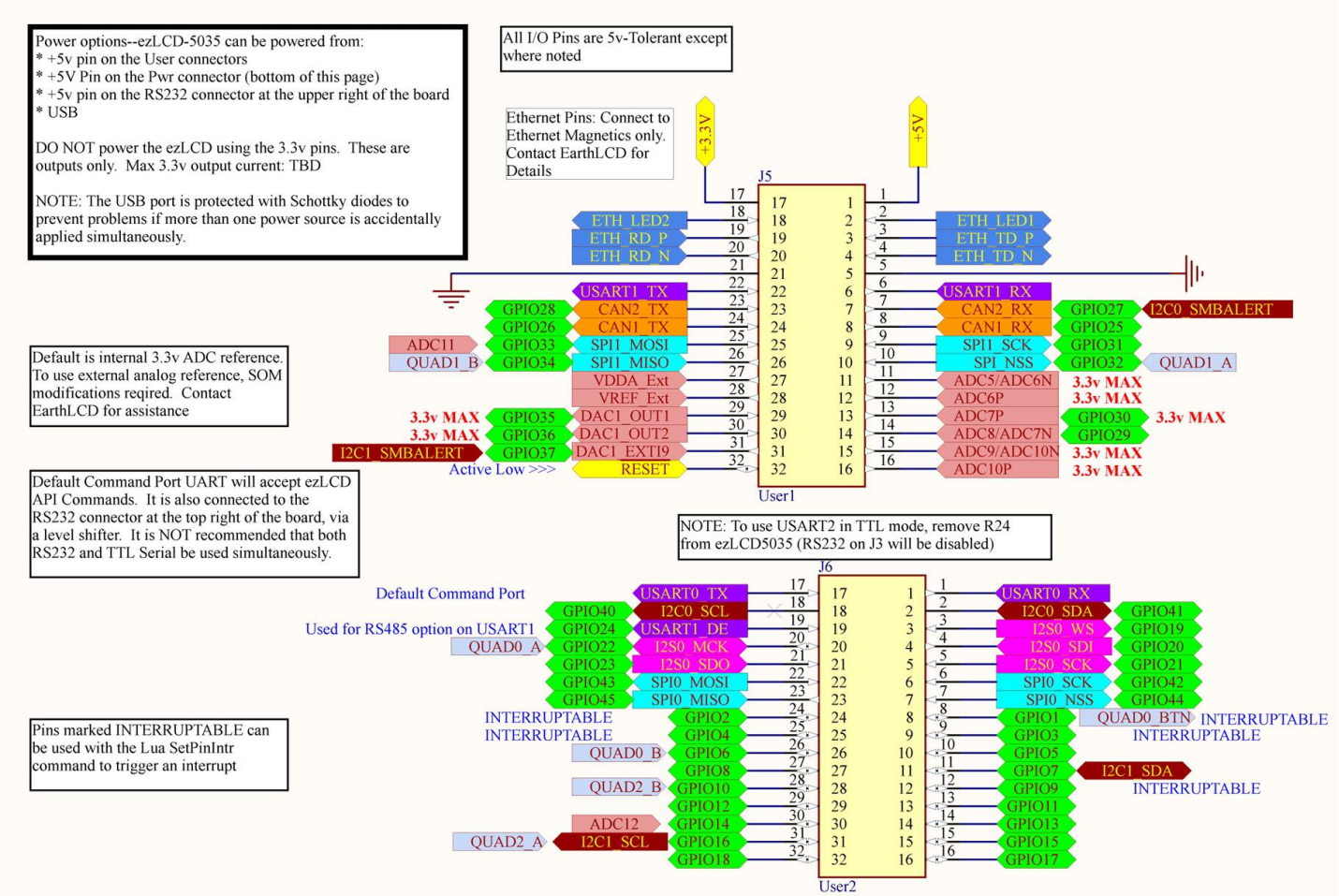
PCB Back



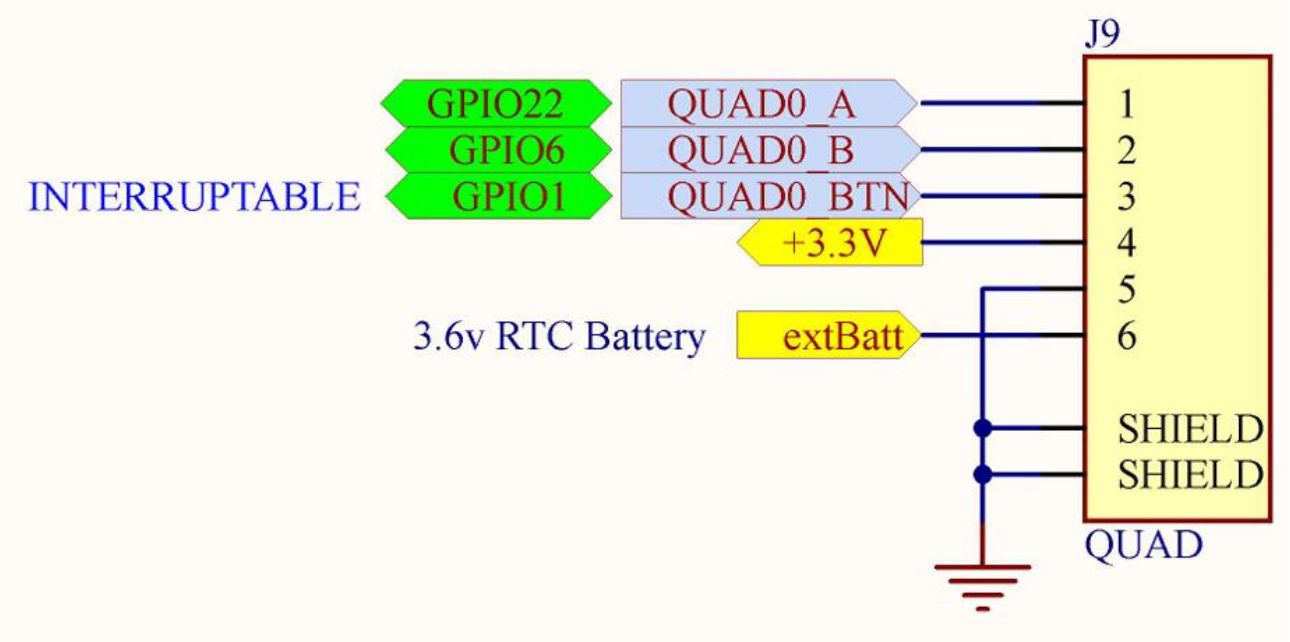
EarthSOM


MicroSD
STM32H757XIH6
DRAM
To remove MicroSD, slide housing down towards latch
EarthSOM-STM32H7
PCB-1901-00-01 REV B
(c) 2021 Earth Computer Technologies, Inc

6.3 J5 & J6 I/O


Power options--ezLCD-5035 can be powered from:
* +5v pin on the User connectors
* +5V Pin on the Pwr connector (bottom of this page)
* +5v pin on the RS232 connector at the upper right of the board
* USB
DO NOT power the ezLCD using the 3.3v pins. These are outputs only. Max 3.3v output current: TBD
NOTE: The USB port is protected with Schottky diodes to prevent problems if more than one power source is accidentally applied simultaneously.
All I/O Pins are 5v-Tolerant except where noted
Ethernet Pins: Connect to Ethernet Magnetics only. Contact EarthLCD for Details
+3.3V
+5V
J5
ETH_LED2
ETH_RD_P
ETH_RD_N
ETH_LED1
ETH_TD_P
ETH_TD_N
USART1_TX
CAN2_TX
CAN1_TX
SPI1_MOSI
SPI1_MISO
VDDA_Ext
VREF_Ext
DAC1_OUT1
DAC1_OUT2
DAC1_EXTI9
RESET
USART1_RX
CAN2_RX
CAN1_RX
SPI1_SCK
SPI1_NSS
ADC5/ADC6N
ADC6P
ADC7P
ADC8/ADC7N
ADC9/ADC10N
ADC10P
GPIO28
GPIO26
GPIO33
GPIO34
GPIO35
GPIO36
GPIO37
GPIO27
GPIO25
GPIO31
GPIO32
GPIO30
GPIO29
ADC11
QUAD1_B
I2C1_SMBALERT
Active Low >>>
I2C0_SMBALERT
QUAD1_A
3.3v MAX
Default is internal 3.3v ADC reference. To use external analog reference, SOM modifications reqired. Contact EarthLCD for assistance
Default Command Port UART will accept ezLCD API Commands. It is also connected to the RS232 connector at the top right of the board, via a level shifter. It is NOT recommended that both RS232 and TTL Serial be used simultaneously.
NOTE: To use USART2 in TTL mode, remove R24 from ezLCD5035 (RS232 on J3 will be disabled)
J6
Default Command Port
Used for RS485 option on USART1
USART0_TX
I2C0_SCL
USART1_DE
I2S0_MCK
I2S0_SDO
SPI0_MOSI
SPI0_MISO
USART0_RX
I2C0_SDA
I2S0_WS
I2S0_SDI
I2S0_SCK
SPI0_SCK
SPI0_NSS
GPIO40
GPIO24
GPIO22
GPIO23
GPIO43
GPIO45
GPIO41
GPIO19
GPIO20
GPIO21
GPIO42
GPIO44
QUAD0_A
INTERRUPTABLE
INTERRUPTABLE
GPIO2
GPIO4
GPIO6
GPIO8
GPIO10
GPIO12
GPIO14
GPIO16
GPIO18
GPIO1
GPIO3
GPIO5
GPIO7
GPIO9
GPIO11
GPIO13
GPIO15
GPIO17
QUAD0_BTN
QUAD0_B
QUAD2_B
ADC12
QUAD2_A
I2C1_SCL
I2C1_SDA
Pins marked INTERRUPTABLE can be used with the Lua SetPinIntr command to trigger an interrupt
User1
User2

6.4 J9 Quadrature Encoder



Pin #	Signal Name	Type	Description
1	QUAD0 A	Input	3.3V Quadrature Phase A Input
2	QUAD0 B	Input	3.3V Quadrature Phase B Input
3	QUAD0 BTN	Input	3.3V Quadrature Push Button Input
4	3.3V	Output	3.3V Quadrature Power
5	GND	Ground	GND Quadrature Power
6	EXT BATT	Input	3.6V RTC Battery

6.5 J8 QWIIC I2C

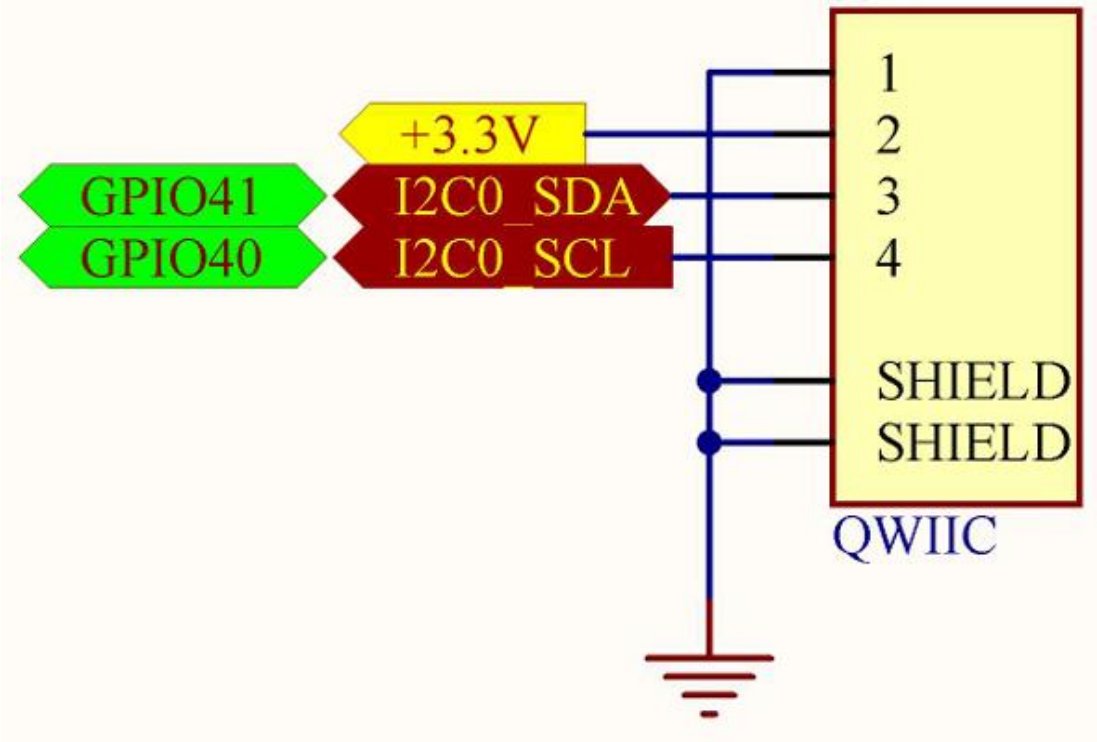
CONNECTOR USED: JST SM04B-SRSS-TB(LF)(SN)

CONNECTOR MATE: JST SHR-04V-S

CONNECTOR CONTACT:

CABLE ASSEMBLY: SparkFun P/N PRT-17260

This versatile connector popularized by SparkFun (QWIIC) and AdaFruit (Stemma QT) connects to 100's of i2c breakout boards with sensors and actuators of all types.

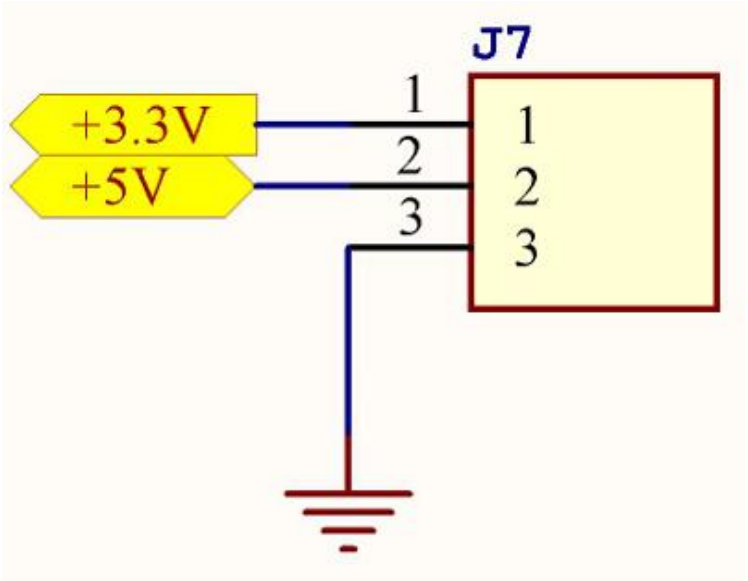


Pin #	Signal Name	Type	Description
1	GND	Ground	
2	3.3V	Output	
3	I2C0 SDA	Output	I2C Data
4	I2C0 SCL	Output	I2C Clock

6.6 J7 DC Output

Located to the lower left of User2 connector (right side of board). It is recommended, if used, to draw no more than 100 milliamps from these pins total.

Connector Used: 3 Position 2.54mm (.1”) Pin header Samtec TSW-103-07-G-S



Pin #	Signal Name	Type	Description
1	3.3V	Output	+3.3 Volt power
2	5V	Output	+5.0 Volt power
3	GND		Ground

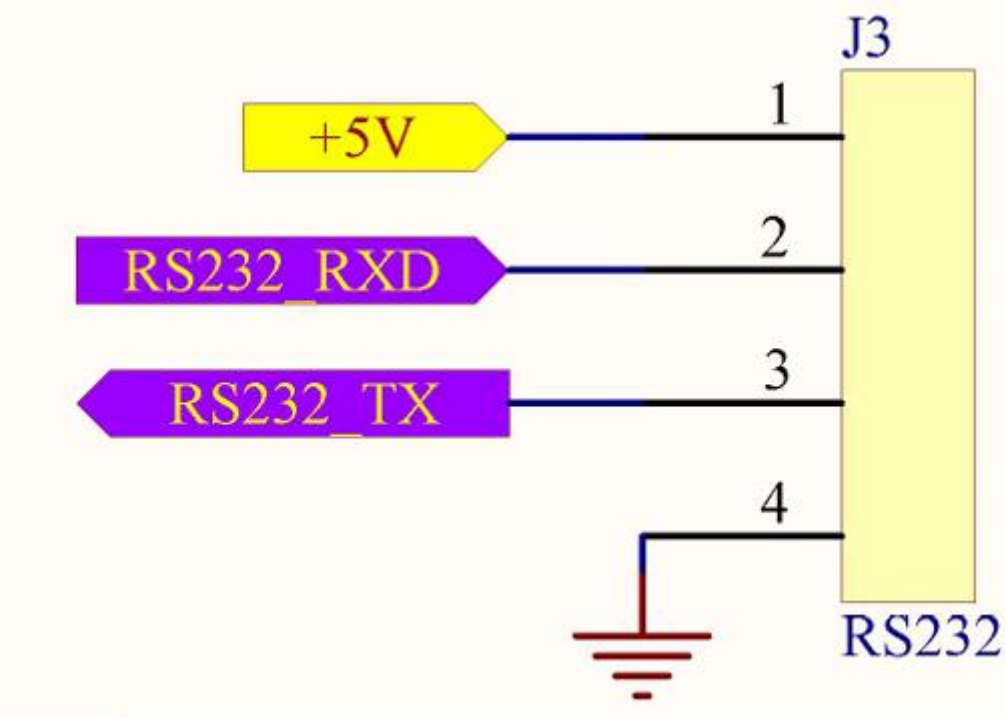
6.7 J3 RS232

Connector Used: TE 292175-4

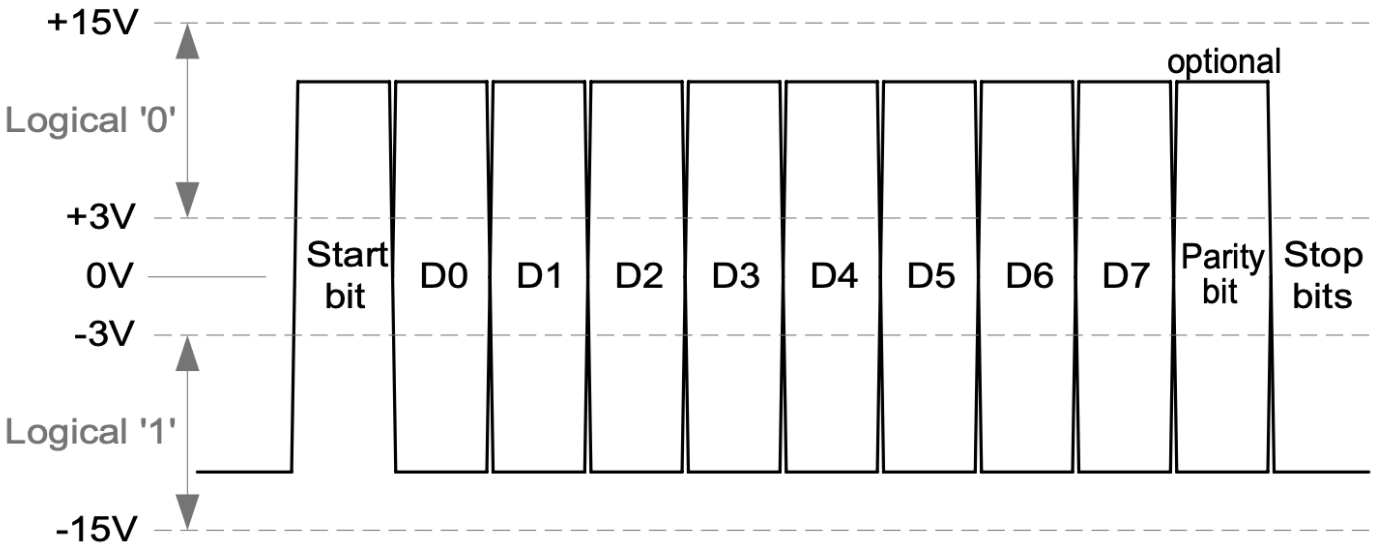
Mating Connector: TE 175778-4

Connector Pins: TE 280708-2

Cable Assembly: TE 2405869-4 (300mm with J3 Mate on both ends!)



Signal Name	Type	Description
+5V	Power Input	
RS232 RX	Discrete Input (-15 to +15)	RS-232 Received Data Input
RS232 TX	Discrete Output (-3 to +3)	RS-232 Transmitted Data Output
GND	Ground	Signal Ground

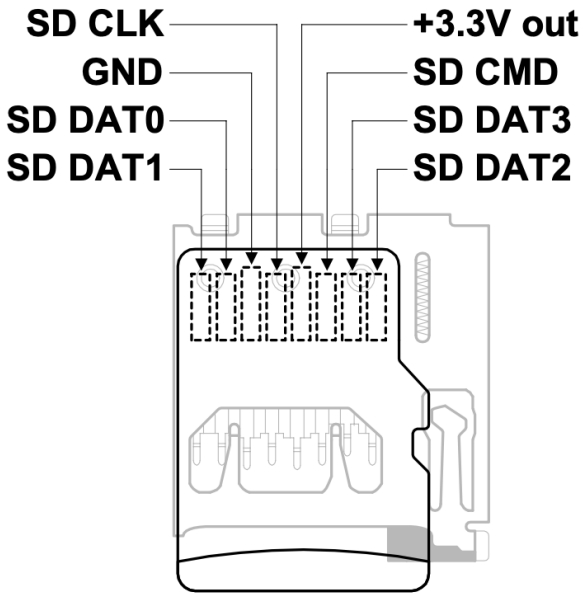

+15V
Logical '0'
+3V
0V
-3V
Logical '1'
-15V
Start bit
D0
D1
D2
D3
D4
D5
D6
D7
optional
Parity bit
Stop bits

6.8 MicroSD

Connector Used: Molex 47219-2001

The Micro SD interface hardware supports the SD Memory Card Specification. SD Memory Card operations are supported by the firmware.

Important Note: You should not have to remove the SD card to delete the autoexec.lua. You can access the ezLCD Boot Options Menu by holding the B1 button down and pressing and releasing the reset button. Then press 'No Autoexec' and then press 'Close'. You can now change the name of the autoexec.lua (or delete it).

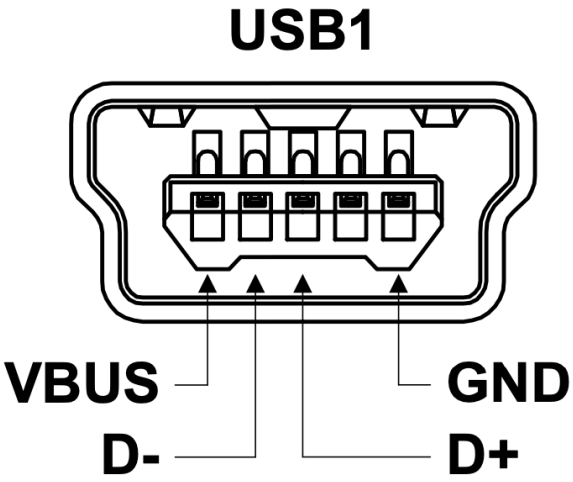


Signal Name	Type	Description
GND	Ground	Common power return and signal ground
+3.3V Out	Pwr/Out	+3.3V/0.5A regulated voltage output. May be used as a power supply for external devices.
SD DAT1	Discrete I/O (0-3.3V or Opened)	SD Data 1 Signal This signal is pulled to +3.3V by an internal resistor
SD DAT2	Discrete I/O (0-3.3V or Opened)	SD Data 2 Signal This signal is pulled to +3.3V by an internal resistor
SD DAT3	Discrete I/O (0-3.3V or Opened)	SD Data 3 Signal

		This signal is pulled to +3.3V by an internal resistor
SD CMD	Discrete output (0-3.3V or Opened)	SD Command Signal This signal is pulled to +3.3V by an internal resistor
SD DAT0	Discrete I/O (0-3.3V or Opened)	SD Data 0 Signal This signal is pulled to +3.3V by an internal resistor
SD CLK	Discrete output (0-3.3V or Opened)	SD Clock

6.9 USB1

Connector Used: Type USB Mini - Molex 513870578

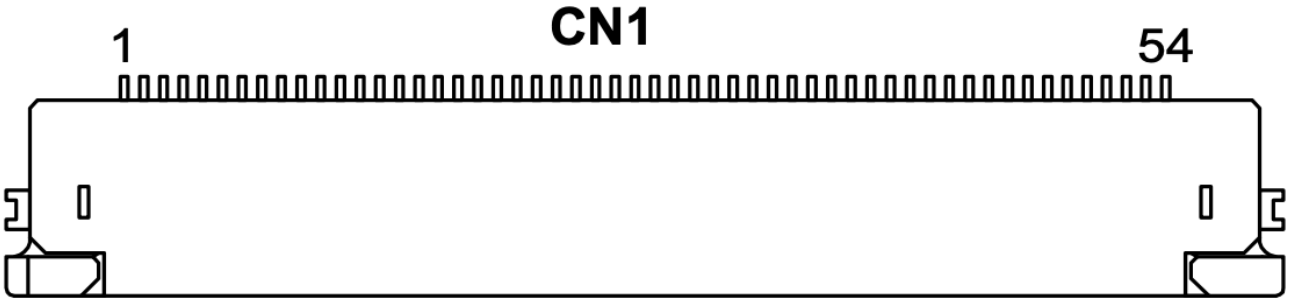


For the detailed description of the USB signals, please refer to the USB 2.0 specification.

The ezLCD-5035-RT USB interface is compatible with the USB 2.0 specification and supports Full Speed (12 Mbps) communication.

6.10 CN1

Connector Used: Kyocera AVX 046240054005800+



Pin No	Name	Type	Description
1,2	LED-	20mA current source	Backlight current source
3,4	LED+		
5,6,7	N/C		Not connected
8	Ty-	Resistance	Touch screen y- input
9	Tx+	Resistance	Touch screen x+ input
10	Ty+	Resistance	Touch screen y+ input
11	Tx-	Resistance	Touch screen y+ input
12-19	B0-B7	Discrete Output (0-3.3V)	8 Bits of RGB Blue component
20-27	G0-G7	Discrete Output (0-3.3V)	8 Bits of RGB Green component
28-35	R0-R7	Discrete Output (0-3.3V)	8 Bits of RGB Red component
36	HSYNC	Discrete Output (0-3.3V)	Horizontal Synchronization
37	VSYNC	Discrete Output (0-3.3V)	Vertical Synchronization
38	DCLK	Discrete Output (0-3.3V)	Data Clock
39,40	N/C		Not connected
41,42	+3.3V Out	Pwr/Out	+3.3V/0.5A regulated voltage output
44-47	N/C		Not connected
48	GND	Gnd	Common power return and signal ground
49,50,51	N/C		Not Connected
52	ENB	Discrete Output (0-3.3V)	Data Enable

53,54	GND	Gnd	Common power return and signal ground

7. Boot Options

The ezLCD-5035-RT ships with a few different boot options:

7.1 Default Startup Screen

If there is no autoexec.lua file, the ezLCD-5035-RTwill load into the Default Startup Screen which features the model, firmware version, and a QR code that links to the serial number of the processor chip.



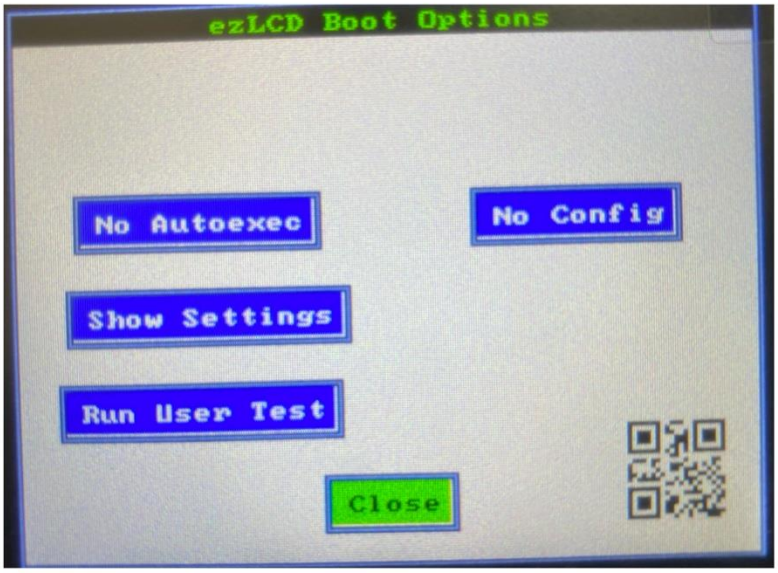
7.2 autoexec.lua

As previously discussed, if any lua file called autoexec.lua is placed in the ezLCD-5035-RT root folder, that script will run on startup.

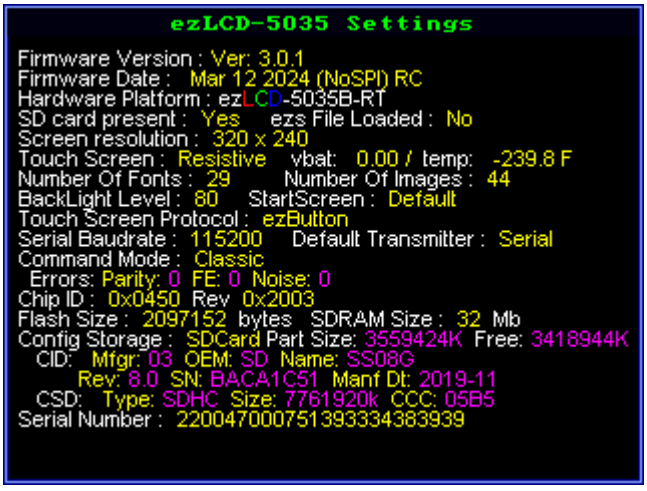
7.3 Boot Options

To enter ezLCD Boot Options, press and hold the B1 button then press and release the reset button. Then release the B1 button and your device should load into the ezLCD Boot Options mode.

ezLCD Boot Options gives the user a few different options as shown below.



- The “No Autoexec” and “No Config” buttons can be toggled on, and after tapping “Close”, will boot the ezLCD-5035-RTwithout the autoexec file or config.txt file respectively on this boot.
    - Note: The yellow text shows that “No Autoexec” is toggled on.
- The “Show Settings” button will load a screen with various boot settings for the device as shown below.



- “Run User Test” will run “usertest.lua” in the root of your ezLCD SD card. NOTE: The “Run User Test” button will only be shown if the usertest.lua is present in the root of the ezLCD File System. The ezLCD-5035-RT ships with a user test program that allows the user to calibrate the screen and will cycle through test scripts that help the user detect visual discrepancies. A

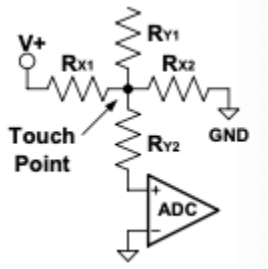
customer might use this functionality to have their own usertest.lua for machine configuration, test, calibration or setup in their embedded product.

8. Touch Screen

8.1 Introduction

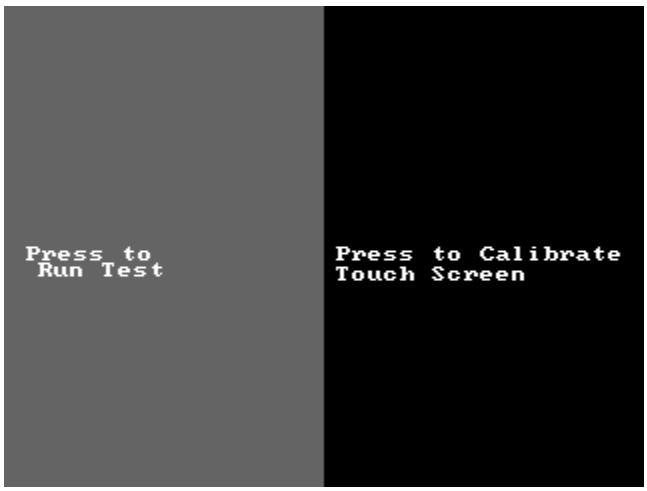
The ezLCD-5035-RT has a 4-wire resistive analog touch screen. This touch screen consists of two layers of transparent resistive material with silver ink for electrodes. These two layers are stacked on an insulating layer of glass, separated by tiny spacer dots. They are interfaced electrically to the dedicated ezLCD-5035-RTA/D converter. During measurement of a given coordinate, one of the resistive planes is powered along its axis and the other plane is used to sense the location of the coordinate on the powered plane.

For example, in the case of the X coordinate measurement, the X plane is powered, as shown on the drawing below. The Y plane is used to sense where the pen is located on the powered plane as follows: At the location where the pen depresses the touch screen, the planes are shorted. The voltage measured on the sensed plane is proportional to the location of the touch on the powered plane. This voltage is then converted by the dedicated ezLCD-5035-RTADC as shown on the drawing below.



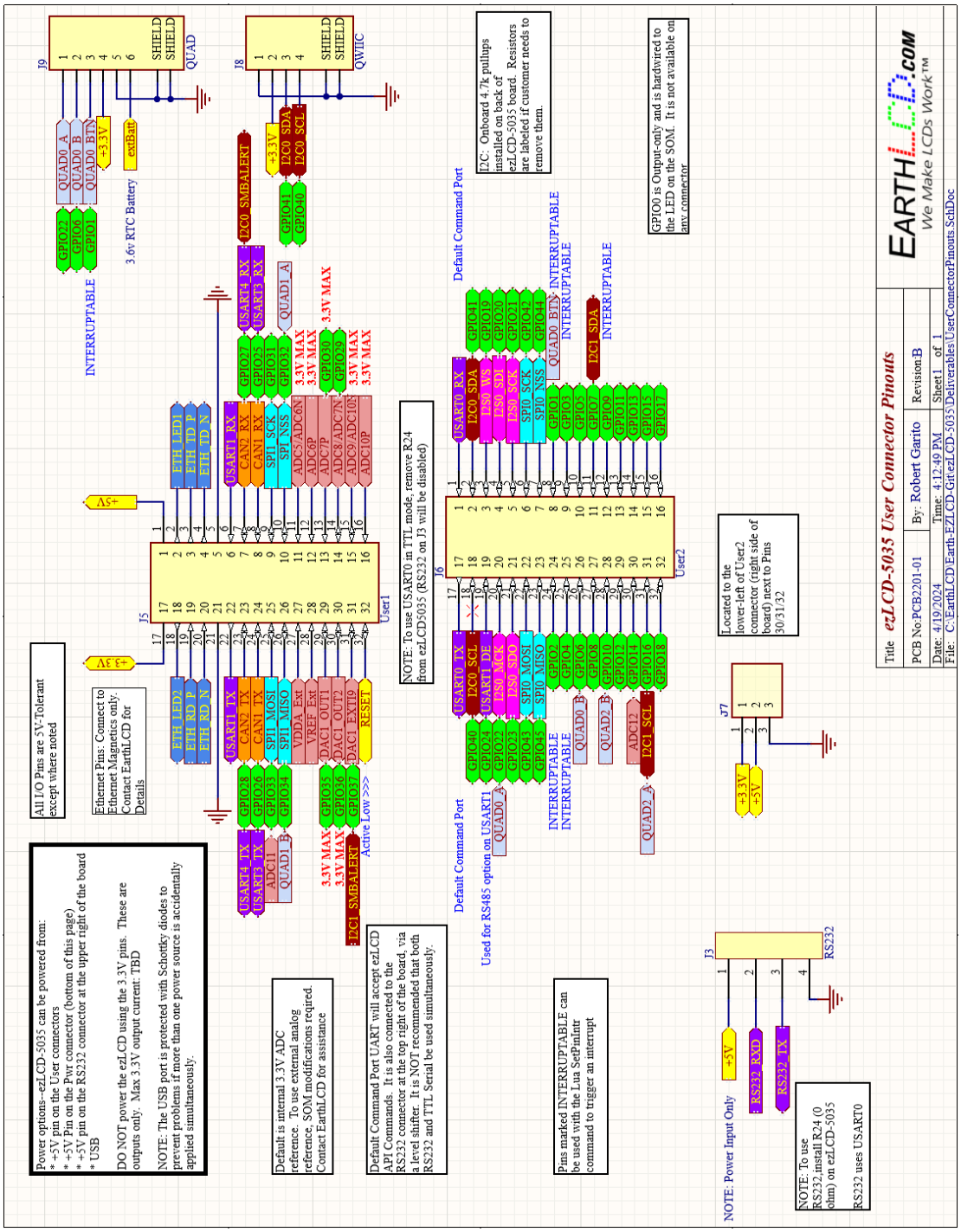
8.2 Calibration

The touch screen calibration is a part of the previously mentioned Boot Options menu. When in Boot Options, press “Run User Test” and the User Test script will run. When prompted, press on the button for calibration, and the calibration procedure will follow.



Note: If User Test is not in the boot menu you can find a copy of it in the \lua directory of the ezLCD Flash Drive. You can also send the calibrate command through the notepad++, or ezLCD Studio program.

Appendix A. User Connector Pinouts



Appendix B. ezLCD Hardware & Software Roadmap

This appendix is to share with potential and current ezLCD customers our plans for expanding both the software API and additional hardware models of the ezLCD-5x . The ezLCD-5x hardware platform is very modular as most models will use the EarthSOM-STM32H7 STM32H757XIH6 microcontroller-based system on module. We list these upcoming models and features to let you know if we have something you might use in your own products. Please contact your EarthLCD sales representative with questions about the timing of the release of this product. And keep in mind we offer engineering, programming and customization services. Also note some of these are already added just awaiting testing, revision, application notes and/or documentation.

Software - Development Tools

- MicroPython
- ezLCD Studio add intellisense for ezLCD commands, debug breakpoints

Software – Command Port expansion

- SPI
- CAN Open

Software – API Commands

- Ethernet

Software – Protocols

- ModBus
- DMX512
- TCP/IP
- CAN
- i2s Digital Audio

Hardware – Development Tools

- ezLCD-5x-EDK a breakout board adding transceivers for CAN Bus & RS-485, ethernet jack, Microphone, Pi Pico footprint, and terminal blocks.

Hardware – New ezLCD Models with different screens

- ezLCD-5035W An 3.5 inch thin ezLCD for badge and electronic label applications
- ezLCD-5031 A 3.1” 720x720 Sunlight Readable CANBus centric ezLCD for Automotive applications
- ezLCD ezLCD-5080-RT An 8 inch 800x600 ezLCD
- ezLCD-5104-RT A 10.4 inch 800x600 ezLCD
- ezLCD-5121-RT A 12.1 inch 1024x768 ezLCD

Application Notes and Documentation:

- Using a Host Processor & the ezLCD API
- Using a Pi Pico as the Host Processor
- Using a Pi Pico W as a Wi-Fi or BT Gateway
- Optional pre-configured STM32Cube Development environment
- EarthSOM-STM32H7 Pinout
- Troubleshooting Guide
- Building an Audio Spectrum Analyzer
- Building a Galatic Weight Scale
- Serial Communications